The Dyson Equation

for Random Matrices

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What is a random matrix?

A *random matrix* is a matrix whose entries are random variables.

$$M = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1n} \\ X_{21} & X_{22} & \cdots & X_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nn} \end{pmatrix}$$

What properties are actually predictable?

We mainly study **eigenvalues** of real symmetric matrices, since

1. They are mathematically tractable.
2. They summarize the large-scale behaviour of the matrix.
3. They appear in many applications (physics, ML, networks).

Spectral Laws

Definition 1

Let $X \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$. The **empirical spectral distribution** of X is the measure

$$\mu_n(A) = \frac{\#\{\lambda_i \in A\}}{n}, \quad A \subseteq \mathbb{R}.$$

If X is a random matrix, μ_n is a random measure.

Main Questions:

- When does $\mu_n \to \mu$ deterministically as $n \to \infty$?
- When it does, what is the distribution of μ?
- Do the distributions of individual entries matter?

Wigner Matrices

- $X \in \mathbb{R}^{n \times n}$ symmetric
- X_{ij} are independent and identically distributed (for $i \leq j$)
- $\mathbb{E}\left[X_{ij}\right] = 0$ and $\mathrm{Var}\left(X_{ij}\right) = \frac{1}{n}$

Examples:

- $X_{ij} \sim \mathcal{N}\left(0, \frac{1}{n}\right)$
- $X_{ij} \in \left\{-\frac{1}{\sqrt{n}}, +\frac{1}{\sqrt{n}}\right\}$, each with probability $1/2$
- $X_{ij} = \frac{1}{\sqrt{n}}\left(Y_{ij} - 1\right)$, where $Y_{ij} \sim \mathrm{Exp}(1)$

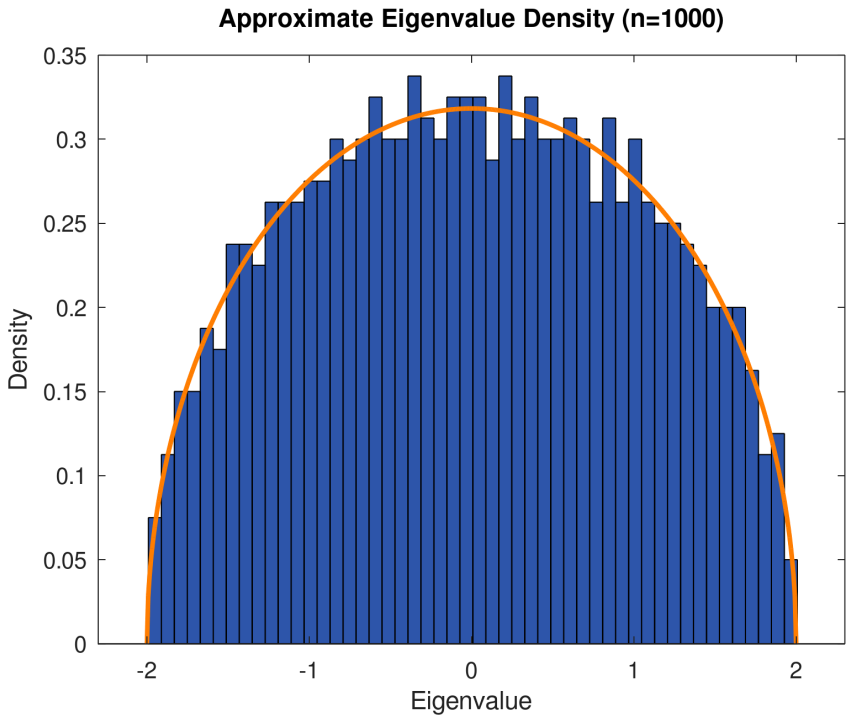
Semicircle Distribution

For **all** Wigner matrices,

$$\mu(A) = \int_A \rho_{\text{sc}}(x)\, dx,$$

where

$$\rho_{\text{sc}}(x) = \frac{1}{2\pi}\sqrt{4 - x^2}, \quad x \in [-2, 2].$$



Important Tools

Definition 2

The **Stieltjes transform** of a probability measure μ is given by

$$m(z) = \int_{\mathbb{R}} \frac{1}{x - z} \, d\mu(x), \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Intuitively, this gives a complex-valued function that encodes μ.

The Stieltjes transform can be inverted to recover μ using

$$\frac{1}{\pi}\lim_{\varepsilon\to 0}\int_a^b \operatorname{Im}\, m(x+i\varepsilon)\,dx = \mu(a,b) + \frac{1}{2}\mu\{a\} + \frac{1}{2}\mu\{b\}.$$

This gives that $\mu_n \to \mu \Leftrightarrow m_n \to m$.

Definition 3

Let $X \in \mathbb{R}^{n \times n}$. For $z \in \mathbb{C}$ such that z is not an eigenvalue of X, the **resolvent** of X is defined as

$$G(z) = (X - zI)^{-1}.$$

If X is a random matrix, $G(z)$ is a random matrix.

Important Tools

The resolvent of X is related to the ST of the ESD of X:

$$\frac{1}{n}\operatorname{tr}(G(z)) = m_n(z)$$

We can now work with the matrix directly to determine limiting μ.

The Dyson Equation

The Dyson Equation

1. *Analyze $G(z)$ using various matrix/probabilistic techniques:*

- Schur complement, independence, Cauchy interlace theorem

$$\frac{1}{n}\operatorname{tr}(G(z)) \doteq -\frac{1}{z+\frac{1}{n}\operatorname{tr}(G(z))+\text{error}}$$

equivalently,

$$m_n(z) \to -\frac{1}{z+m_n(z)}, \quad \text{or} \quad 1+(z+m_n(z))\cdot m_n(z) \to 0$$

2. *Perform stability analysis*

- Show "equation converges" $\Rightarrow$ "m_n converges to true solution"

$$|1 + (z + m_n(z)) \cdot m_n(z)| < \varepsilon$$
$$\overset{\cdots}{\Rightarrow} |m_n(z) - m(z)| < C \cdot \varepsilon$$

where $m(z)$ satisfies

$$1 + (z + m(z)) \cdot m(z) = 0$$

This is the **Dyson Equation** for Wigner matrices.

3. *Solve for m and invert to recover μ*

$$1 + (z + m(z)) \cdot m(z) = 0$$

$$\Rightarrow m(z) = \frac{-z + \sqrt{z^2 - 4}}{2}$$

$$\Rightarrow \mu(a, b) = \int_a^b \underbrace{\frac{1}{2\pi}\sqrt{4 - x^2}}_{\text{semicircle density}} \, dx$$

Done! A similar process can be used for other types of matrices.

Main Questions:

When does $\mu_n \to \mu$ deterministically as $n \to \infty$?

Among other cases, the Wigner Ensemble.

When it does, what is the distribution of μ?

The semicircle distribution.

Do the distributions of individual entries matter?

No!

L. Erdős, **The Matrix Dyson Equation and Its Applications for Random Matrices**, arXiv:1903.10060 (2019).

E. Paquette, **Random Matrix Theory and Optimization**, lecture notes, CRM-PIMS Summer School, 2024. Available at: https://elliotpaquette.github.io/notes/RMTandOptimization-20240701.pdf